

McHenry, IL 60051
312.436.1932
pbirmingham@gmail.com
[LinkedIn](#)

PHILLIP J. BIRMINGHAM

CAREER PROFILE

Experienced machine learning engineer with demonstrated experience in machine learning and software engineering across insurance, banking, and consulting industries.

Core Competencies Include: Advanced Analytics, Applied ML & Data Science, Automation & AI, Cloud infrastructure, Containerization, Cross-functional Collaboration, Data engineering, Machine learning, ML Software Engineering, Python Software Development, Ruby Platform Expertise, Software Development

EXPERIENCE

Independent Contractor, Mercor Intelligence - Remote

May 2026 - Present

- Provided expertise in data science and software engineering to improve models for a top AI lab.

Senior Data Scientist, Vericity - Remote

Apr 2020 - Dec 2025

- Developed machine learning models for life insurance applicant behavior and retention using Python, pandas, scikit-learn, and XGBoost, serving thousands of daily users.
- Researched, tested, and deployed when-to-call models that increased call-center contact rates by up to 10%.
- Researched and developed models to predict long-term value of sales leads pre-purchase, enabling more informed buying decisions.
- Implemented dynamic bidding subsystem for lead purchases that optimized acquisition efficiency.
- Analyzed effectiveness of voicemail messaging used by sales call center to improve outreach strategy; recommended a single message to avoid do-not-call requests.
- Developed model predicting life-insurance applicant follow-through to payment using historical applicant behavior, enabling marketers to prioritize follow-up outreach.
- Presented when-to-call models, locally-hosted LLMs, and retrieval-augmented generation to the research and technology department and CTO, rapidly mastering RAG; motivated team experimentation with LLM technology.
- Created infrastructure-as-code for deployed machine learning projects to support scalable operations.
- Implemented PowerBI report to automate querying policy status by retention team, reducing time to get policy status by over half.
- Engineered prompts for LLM-powered code review application that won the company-wide hackathon.
- Implemented SonarCloud security recommendations in CI/CD pipeline, systematically hardening code and reducing vulnerability findings in scans.
- Collaborated with security team to encrypt PII storage in S3, reducing security audit findings.
- Engineered Python training and inference pipelines for retention when-to-call and applicant payment models with comprehensive unit tests using fixtures and mocks to isolate model logic, enabling teammate contributions and production deployment without breaking changes.

- Queried and joined 2,300-column dataset across 12+ database tables with SQL to create reusable feature store for model development, enabling when-to-call retention model training and testing at scale.
- Applied life insurance domain knowledge of policy termination sequences to define payment model targets, distinguishing final status changes from intermediate events to prevent training data misclassification and improve model accuracy.
- Partnered with Workforce Management to pause when-to-call model rollout by demonstrating that proposed A/B test populations were not equivalent, protecting decision quality and preserving model credibility for properly-designed future experiment.
- Built Docker images for AWS Batch, containerizing ML inference workloads to enable automated batch inference at scale.

Senior Software Engineer, Vericity - Remote

Oct 2014 - Apr 2020

- Developed full stack and distributed applications in Ruby, React, and TypeScript serving hundreds of daily users across life insurance policy and billing workflows.
- Fixed resource leak in open-source RPC library for RabbitMQ by implementing proper connection cleanup logic, preventing connections from remaining open unnecessarily.
- Updated TrustCommerce TCLink gem to work with contemporary versions of the TrustCommerce API, enabling migration from JRuby to MRI Ruby.
- Built core UI and backend services in React and TypeScript with integrated third-party libraries, enabling the company's first B2C sales system.
- Built user-facing payment gateway to process core payment flows for B2C sales system.
- Designed and implemented automated end-to-end test of billing system, enabling validation of core payment flows.
- Designed and implemented document handling and workflow components for microservices-based electronic new business system, enabling migration from legacy monolith.
- Facilitated shift from JRuby to MRI Ruby by modernizing TrustCommerce gem to work with current versions of TrustCommerce API.
- Directed external vendor group across payment processing, document processing, and reinsurance domains by defining detailed specifications upfront, ensuring alignment between external developers and internal team and reducing integration risk.
- Mentored QA lead in Ruby via pair programming, enabling test suite extensions without engineer involvement.
- Demonstrated test-driven development to members of a different team, enabling them to gain the benefits of TDD.

Senior Software Engineer, Pathfinder Software - Chicago, IL

Nov 2013 - Sep 2014

- Developed full stack applications in .NET and AngularJS for a swimming pool equipment manufacturer serving hundreds of daily users.
- Specified hardware system requirements at client company, working with their developers to ensure smooth implementation of new software.
- Coordinated rollout of system testing using SpecFlow, MSpec, Cucumber, and Jasmine.
- Drove system design decisions and implemented core code for remote-control pool filter and heater system, enabling clients to monitor and control equipment via app and web interface.
- Facilitated standups, sprint planning, and sprint reviews, eliminating blockers and keeping team moving forward.
- Analyzed past client request patterns and prepared demonstrating engineer to showcase database tables at sprint demonstration, reducing response time and increasing client satisfaction.
- Walked client through websocket vs. polling tradeoff analysis, earning approval for chosen architecture at project onset.
- Led four-person team through code review and testing practice example, reducing bugs across implementations.

Senior Software Engineer, Independent Contractor - Remote

May 2002 - Sep 2013

- Developed desktop applications and libraries in .NET, Ruby, and Java serving hundreds of daily users across entrepreneurial, educational, and precision agriculture sectors.
- Designed and developed class library to perform interpolation of measurement over 2D grid supporting kriging with spherical and exponential variogram models.
- Designed and developed desktop application to conduct enrollment lottery for Los Angeles-area charter school using Visual Basic .NET and Microsoft Excel/SQL Server Data Engine.
- Designed jellybean mosaic art system with color mapping and rendering logic, enabling entrepreneur to pitch working prototype to candy company.
- Built custom SQL Server database interface for MLS listings, enabling faster retrieval and local data access for real estate operations.
- Developed interface to manage customer records by integrating with existing IBM database, enabling quick view, edit, and add operations that improved data accuracy.
- Built email order processing application using Ruby and IMAP to extract Americart orders, reducing manual entry time and errors.
- Built IMAP mail parser extracting order information from unstructured email into MySQL database schema, enabling reporting without manual data entry.
- Evaluated chunk sizes and indexing strategies for Logseq RAG retrieval, improving downstream answer accuracy through systematic tuning against fixed embedding model.

EDUCATION

Vanderbilt University, Nashville, TN

Master of Science - Physics

Arkansas State University, Jonesboro, AR

Bachelor of Science - Physics

Georgia Institute of Technology, Atlanta, GA

Master of Science - Computer Science (Interactive Intelligence)

SKILLS

AI & Machine Learning, Predictive Modeling, Software Architecture & Design, Systems Integration, Data Pipeline Development, Agile & Scrum Development, Technical Leadership & Mentoring, Financial Services Experience, Python, Go, Ruby, Java, Javascript, C, C++, Linux, Windows, MacOS, Jupyter, XGBoost, Optuna, Git, Atlassian, Visual Studio Code, Terraform, Ollama, Anthropic Claude, Power BI, pandas, LLM, LangChain, RAG, Amazon Redshift, PostgreSQL, MySQL, Microsoft SQL Server